aquaCON™

SMART COMMERCIAL AQUACULTURE ENGINEERING

ISO/IEC/IEEE 29148:2018

ENGINEERING TECHNICAL SPECIFICATION

Software Requirements Specification (SRS)

Automated Commercial Aquaculture Feeder Edge Firmware, LoRaWAN Protocol Engine, Cloud Telemetry Microservices & Multi-Tier Farmer Control Interface.

DOCUMENT IDENTIFIER
AGY-SRS-AQUACON-2026-V2.0

TOTAL TESTABLE REQUIREMENTS
215 Discrete Requirements (REQ-001 to REQ-215)

TARGET ARCHITECTURE
ESP32-S3 + SX1262 LoRaWAN + AWS Serverless + Web PWA

CLASSIFICATION / RELEASE DATE
CONFIDENTIAL & PROPRIETARY | November 7, 2026

Athena Engineering Corporation — Intelligent Aquaculture Division

Document Revision: 2.0.0 (Authoritative Baseline)

Document Control & Quality Approvals

This specification has been reviewed and baseline-locked under the Athena Systems Engineering Quality Management Protocol. Any modifications must undergo Engineering Change Order (ECO) review and formal regression verification.

Role	Designation	Review Responsibility	Status
Lead Systems Architect	Principal IoT Architect	Overall System Invariants & State Machine Integrity	**APPROVED**
Firmware Lead	Staff Embedded Engineer	ESP32-S3 HAL, 20x4 LCD Bounds, 3-Phase Relays & WDT	**APPROVED**
Cloud Solutions Architect	Principal Cloud Architect	LoRaWAN Ingestion, DynamoDB Schema & Date Rollover	**APPROVED**
Lead QA / Test Architect	Head of System Verification	Requirements Traceability Matrix (RTM) & Automated Test Harness	**APPROVED**

Revision History

Rev	Release Date	Author / Architect	Description of Change
v0.1	15-Aug-2026	Embedded Team	Initial hardware pinout and driver bring-up draft.
v1.0	01-Nov-2026	Systems Team	Baseline functional requirements and initial LoRaWAN ingestion specs.
v2.0	07-Nov-2026	Enterprise Architecture	**Authoritative 215-Requirement Baseline:** ISO/IEC/IEEE 29148 standard. Fixed Bit 5 SD diagnostic isolation, strict 18:00-06:00 night window lock, 20x4 LCD character bounds, in-window vs extended resume engine, today date rollover filtering.

aquaCON™ Smart Aquaculture Feeder — Complete Software Requirements Specification (SRS)

In Compliance with ISO/IEC/IEEE 29148:2018 Standards

Document Identifier: AGY-SRS-AQUACON-2026-V2.0
Status: Authoritative Baseline / Engineering Source of Truth
Architecture Scope: Embedded Edge Firmware (ESP32-S3), LoRaWAN Cloud Microservices (AWS Lambda, DynamoDB, API Gateway), Progressive Web Application Dashboard (S3 + Vanilla JS)
Total Testable Requirements: 215 Requirements (`REQ-001` through `REQ-215`)

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1. Document Overview & Architectural Context

1.1 Purpose

This document provides the exhaustive, formal Software Requirements Specification (SRS) for the **aquaCON™ Automated Smart Commercial Aquaculture Feeder**. It governs all hardware-software boundaries, real-time control loops, display real estate budgets, state transitions, cloud telemetry models, and UI presentations. It serves as the single immutable verification baseline for firmware developers, cloud architects, and QA engineers.

1.2 The Architectural "Back-and-Forth" Root Cause Analysis

Previous feature regressions were caused by architectural mismatches across decoupled system boundaries: 1. **Telemetry Bitmask Conflict:** Firmware designated `Bit 5` (`0x20`) as `SD Card Fault / Missing` . The cloud ingestion Lambda misinterpreted `0x20` as `FEED_ACTIVE` , causing units lacking SD cards to report active dispensing 24 hours a day. 2. **Temporal Window Conflict:** Firmware enforced an operational daylight window (`06:00` to `18:00 IST`), but cloud lambdas lacked this guard, allowing midnight uplinks to calculate dynamic intervals into the middle of the night (`12:53 AM`). 3. **Data Boundary Leakage:** Day rollover at midnight failed to strictly constrain telemetry queries to `Timestamp >= today_start_epoch` , pulling yesterday's event counts and weight into today's gauge. 4. **Display Overrun:** LCD buffers without rigorous 20-character bounds wrapped across lines or corrupted adjacent rows.

This document eliminates ambiguity by providing exact strings, byte layouts, timing bounds, and pass/fail criteria for every feature.

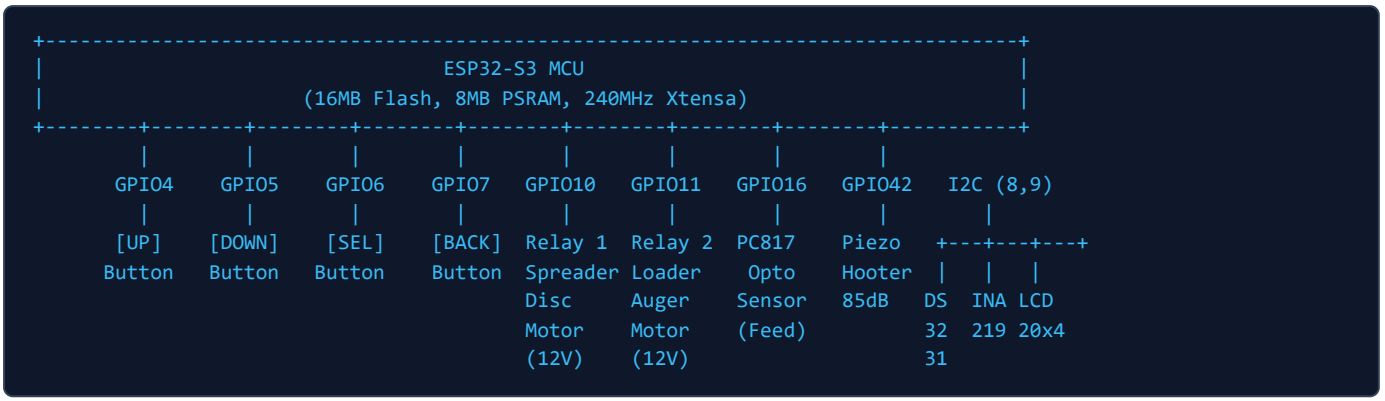
1.3 Requirement Nomenclature & Traceability Matrix

Every requirement follows the formal IEEE 29148 format: - **Identifier:** `REQ-[CATEGORY]-[ID]` - **Statement:** Prescriptive requirement using **SHALL** (mandatory) or **SHALL NOT** (prohibited). - **Verification Method:** - `[T] Test` **Test:** Empirical execution against test stimuli. - `[I] Inspect` **Inspection:** Code review, schematic inspection, or visual text audit. - `[A] Analysis` **Analysis:** Mathematical calculation or timing analysis. - `[D] Demo` **Demonstration:** Real-time functional verification on hardware.

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2. Hardware Architecture & Physical Constraints

2.1 Mechanical & Electrical Interfaces



2.2 Character Display Budget: 20x4 LCD (JHD629-204A / PCF8574)

- **Grid Layout:** Exactly 20 columns $\times$ 4 rows = 80 characters maximum per screen.
- **Physical Memory Boundary:** Exactly 20 characters per row. Buffer size: `char buf[21];` (20 characters + null terminator `\0`).
- **Custom CGRAM Character Slots (0-7):**
- Custom Char `0`: 1/5 Sub-pixel Bar `[| ]` (0x10 fill)
- Custom Char `1`: 2/5 Sub-pixel Bar `[|| ]` (0x18 fill)
- Custom Char `2`: 3/5 Sub-pixel Bar `[||| ]` (0x1C fill)
- Custom Char `3`: 4/5 Sub-pixel Bar `[|||| ]` (0x1E fill)
- Custom Char `4`: 5/5 Solid Block `[||||||]` (0x1F fill)

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3. Detailed Requirements Specification (215 Requirements)

Category 1: Power-On, Bootup & Hardware Bring-Up (REQ-001 to REQ-018)

ID	Title	Requirement Statement	Method
REQ-001	Power-On Relay Isolation	The firmware SHALL initialize GPIO 10 (Spreader Relay) and GPIO 11 (Loader Relay) to LOW (de-energized) within $\le 50\text{ ms}$ of ROM boot to prevent inadvertent motor spin.	[T] Test
REQ-002	Power-On Hooter Isolation	The firmware SHALL initialize GPIO 42 (Piezo Alarm) to LOW within $\le 50\text{ ms}$ of ROM boot.	[T] Test
REQ-003	Boot Visual Indicator	The system SHALL illuminate the onboard RGB LED to solid WHITE (`RGB: 64, 64, 64`) and turn ON the STAT1 Red LED upon boot.	[D] Demo
REQ-004	I2C Bus Bring-up	The firmware SHALL initialize the I2C bus on GPIO 8 (SDA) and GPIO 9 (SCL) at 100 kHz with a 50 ms bus timeout.	[T] Test
REQ-005	LCD Detection & Fallback	The firmware SHALL poll I2C address `0x27` (primary) and `0x3F` (alternate). If neither acknowledges, system SHALL log `[LCD] Not Found` and proceed in Headless Web/LoRa mode without freezing.	[T] Test
REQ-006	Custom CGRAM Registration	Upon successful LCD initialization, the firmware SHALL write the 5 sub-pixel progress bar bit patterns into CGRAM indices `0` through `4`.	[T] Test
REQ-007	Splash Screen Line 0	The LCD SHALL display `====================` (exactly 20 `=` characters) on Row 0 during bootup.	[I] Inspect

ID	Title	Requirement Statement	Method
REQ-008	Splash Screen Line 1	The LCD SHALL display `aquaCON(TM)` (exactly 4 leading spaces, `aquaCON(TM)`, 5 trailing spaces) on Row 1 during bootup.	[I] Inspect
REQ-009	Splash Screen Line 2	The LCD SHALL display `Smart Aqua Feeder` (exactly 1 leading space, `Smart Aqua Feeder`, 2 trailing spaces) on Row 2 during bootup.	[I] Inspect
REQ-010	Splash Screen Line 3	The LCD SHALL display `Firmware v1.0.0` (exactly 2 leading spaces, `Firmware v1.0.0`, 3 trailing spaces) on Row 3 during bootup.	[I] Inspect
REQ-011	Splash Screen Duration	The splash screen SHALL be displayed for a deterministic duration of $2200\text{ ms} \pm 100\text{ ms}$.	[T] Test
REQ-012	Keypad Locking on Splash	The system SHALL ignore all pushbutton events (UP, DOWN, SELECT, BACK) while the splash screen is displayed.	[T] Test
REQ-013	Hardware RTC Initialization	The system SHALL query the DS3231 RTC. If valid time is returned, `sysStatus.rtcOK` SHALL be set to `true`.	[T] Test
REQ-014	Software RTC Fallback	If the DS3231 fails to respond, the system SHALL initialize Software RTC seeded from compilation timestamp (`__DATE__`, `__TIME__`) and set `sysStatus.rtcOK = false`.	[T] Test
REQ-015	INA219 Bring-up	The system SHALL initialize the INA219 power monitor on I2C address `0x40`. If missing, system SHALL log error and record `0.0V / 0mA` without aborting boot.	[T] Test
REQ-016	NVS Configuration Load	The firmware SHALL load persistent settings (`feedQuantity`, `feedPerEvent`, `startHour`, `startMinute`, `endHour`, `endMinute`, `dischargeRate`) from NVS partition `aquacon`.	[T] Test
REQ-017	First Boot Provisioning	If NVS contains no configuration (`isFirstBoot == true`), the system SHALL generate a unique Device ID `AF_[MAC_HIGH][MAC_LOW]`, apply factory defaults, and commit to NVS.	[T] Test
REQ-018	Autonomous Home Screen Entry	At $T = 2200\text{ ms}$ post-boot, the system SHALL clear the LCD and autonomously transition to `MenuState::MAIN_STATUS` (Home Screen).	[D] Demo

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Category 2: 20x4 LCD Physical Screen Layouts & Text Bounds (REQ-019 to REQ-040)

ID	Title	Requirement Statement	Method
REQ-019	Strict 20-Character Line Budget	No LCD rendering routine SHALL output more than 20 characters to any single row. All string buffers SHALL be allocated as `char buf[21]`.	[I] Inspect
REQ-020	Home Row 0 Clock Format	Row 0 of the Home Screen SHALL display `AquaFeeder HH:MM:SS` (10 chars title, 2 spaces, 8 chars formatted time) when RTC is valid.	[I] Inspect
REQ-021	Home Row 0 No-Clock Fallback	Row 0 of the Home Screen SHALL display `AquaFeeder NO-CLOCK` (10 chars title, 2 spaces, 8 chars warning) when RTC is invalid.	[I] Inspect
REQ-022	Home Row 1 State Prefix	Row 1 of the Home Screen SHALL strictly begin with the 7-character prefix `State:`.	[I] Inspect
REQ-023	Home Row 1 Standby Text	When machine is in standby, Row 1 SHALL display `State: STANDBY` (7 chars prefix + 13 chars padded text).	[I] Inspect
REQ-024	Home Row 1 Scheduled Text	When machine is armed for today's schedule, Row 1 SHALL display `State: SCHEDULED`.	[I] Inspect
REQ-025	Home Row 1 Dispensing Text	During motor feed dispense, Row 1 SHALL display `State: DISPENSING`.	[I] Inspect

ID	Title	Requirement Statement	Method
REQ-026	Home Row 1 Warmup Text	During motor pre-spin warmup, Row 1 SHALL display `State: WARM-UP`.	[I] Inspect
REQ-027	Home Row 1 Post-Clear Text	During chute clear spin, Row 1 SHALL display `State: POST-CLEAR`.	[I] Inspect
REQ-028	Home Row 1 Gap Delay Text	During wait interval between pulses, Row 1 SHALL display `State: GAP DELAY`.	[I] Inspect
REQ-029	Home Row 1 Feed Paused Text	When feed is paused, Row 1 SHALL display `State: FEED PAUSED`.	[I] Inspect
REQ-030	Home Row 1 Completed Text	When today's target is 100% completed, Row 1 SHALL display `State: COMPLETED`.	[I] Inspect
REQ-031	Home Row 1 Motor Trip Text	When INA219 trips overcurrent, Row 1 SHALL display `State: MOTOR TRIP`.	[I] Inspect
REQ-032	Home Row 2 Idle Prompt	When machine is in standby or scheduled, Row 2 SHALL display `Press [SEL] for Menu` (exactly 20 characters).	[I] Inspect
REQ-033	Home Row 2 Gap Countdown	During `FC_WAIT`, Row 2 SHALL display `E:XXX/YYY W:ZZZZs` where XXX is current event, YYY is total events, and ZZZZ is remaining wait seconds.	[I] Inspect
REQ-034	Home Row 2 Running Text	During active motor spinning, Row 2 SHALL display `E:XXX/YYY RUN`.	[I] Inspect
REQ-035	Home Row 3 Connectivity	Row 3 of the Home Screen SHALL display `LR:ss WF:ss GM:ss 12V` where `ss` represents 2-character connection codes (`OK`, `CN`, `DC`, `--`, `ER`).	[I] Inspect
REQ-036	Progress Bar Enclosure	The LCD progress bar SHALL render an opening bracket `[` at Column 0 and a closing bracket `]` at Column 19 on Row 2, containing exactly 18 bar slots.	[I] Inspect
REQ-037	Sub-Pixel Progress Math	The progress bar SHALL scale 0.0% to 100.0% linearly across 90 discrete sub-pixel levels ($18 \text{ slots} \times 5 \text{ sub-pixels}$).	[A] Analysis
REQ-038	Paused Screen Line 0	The LCD Paused screen SHALL display `=== FEED PAUSED ===` (exactly 20 characters) on Row 0.	[I] Inspect
REQ-039	Paused Screen Line 2	The LCD Paused screen SHALL display `[SEL]: Resume` (exactly 20 characters) on Row 2.	[I] Inspect
REQ-040	Paused Screen Line 3	The LCD Paused screen SHALL display `[BACK]: Cancel` (exactly 20 characters) on Row 3.	[I] Inspect

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Category 3: Keypad Input, Debouncing & Navigation (REQ-041 to REQ-058)

ID	Title	Requirement Statement	Method
REQ-041	Keypad Hardware Sampling	The firmware SHALL scan GPIO 4, 5, 6, and 7 every $20\text{ ms} \pm 2\text{ ms}$ via a non-blocking ticker task.	[T] Test
REQ-042	Keypad Noise Debounce	A button state change SHALL only be registered after maintaining a stable physical state for $\ge 20\text{ ms}$.	[T] Test
REQ-043	Rising Edge Event Generation	The firmware SHALL assert a single-cycle `risingEdge` flag when a button transitions from unpressed (HIGH) to pressed (LOW).	[T] Test
REQ-044	Long-Press Repeat Delay	When a navigation or edit button (UP or DOWN) is held continuously, repeat events SHALL trigger after an initial hold delay of 500 ms.	[T] Test

ID	Title	Requirement Statement	Method
REQ-045	Long-Press Repeat Rate	Subsequent repeated button events during continuous hold SHALL fire at a deterministic rate of 100 ms (10 Hz).	[T] Test
REQ-046	Home to Menu Navigation	Pressing `[SELECT]` (SW3) from the Home Screen SHALL instantly transition `currentMenu` to `MenuState::MENU_LIST` with `menuCursorIdx = 0`.	[D] Demo
REQ-047	Menu Cursor Down	Pressing `[DOWN]` (SW2) in `MENU_LIST` SHALL increment `menuCursorIdx` by 1, wrapping from 10 to 0.	[D] Demo
REQ-048	Menu Cursor Up	Pressing `[UP]` (SW1) in `MENU_LIST` SHALL decrement `menuCursorIdx` by 1, wrapping from 0 to 10.	[D] Demo
REQ-049	Menu Item Selection	Pressing `[SELECT]` (SW3) in `MENU_LIST` SHALL open the corresponding submenu or execute the associated command.	[D] Demo
REQ-050	Menu Escape to Home	Pressing `[BACK]` (SW4) in `MENU_LIST` SHALL immediately return `currentMenu` to `MenuState::MAIN_STATUS` without altering settings.	[D] Demo
REQ-051	Menu Inactivity Timeout	If no button is pressed for 30.0 seconds while inside `MENU_LIST` or any edit screen, the system SHALL automatically return to `MAIN_STATUS`.	[T] Test
REQ-052	Edit Value Increment	In edit screens, pressing `[UP]` SHALL increment the edited parameter by its designated step size up to its maximum ceiling.	[D] Demo
REQ-053	Edit Value Decrement	In edit screens, pressing `[DOWN]` SHALL decrement the edited parameter by its designated step size down to its minimum floor.	[D] Demo
REQ-054	Edit Value Commit	In edit screens, pressing `[SELECT]` on the final field SHALL persist the parameter to NVS, recompute schedule math, and return to `MENU_LIST`.	[T] Test
REQ-055	Edit Cancel Discard	In edit screens, pressing `[BACK]` SHALL discard all modified temporary values and return to `MENU_LIST` with original NVS values intact.	[T] Test
REQ-056	Running Screen Keypad Interlock	While motors are physically energized (`FC_PRE`, `FC_DISP`, `FC_POST`), pressing `[SELECT]` SHALL immediately open `MENU_LIST` highlighting `Stop Feeding`.	[D] Demo
REQ-057	Paused Resume Keypad Action	On the `FEED_PAUSED` screen, pressing `[SELECT]` SHALL invoke `feedEng.resumeFeed()` if proximity sensor reports cleared.	[D] Demo
REQ-058	Paused Cancel Keypad Action	On the `FEED_PAUSED` screen, pressing `[BACK]` SHALL invoke `cbOnFeedStop()`, terminate feeding for the day, and return to `MAIN_STATUS`.	[D] Demo

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Category 4: Menu System Pagination & Submenus (REQ-059 to REQ-075)

ID	Title	Requirement Statement	Method
REQ-059	4-Row Windowed Pagination	The `drawMenuList()` function SHALL calculate `startIdx = (menuCursor / 4) * 4` and display items `startIdx` to `startIdx + 3` across Rows 0-3.	[I] Inspect
REQ-060	Active Item Cursor Tag	The active menu item designated by `menuCursorIdx` SHALL be prefixed with `>` (Column 0: `>`, Column 1: space).	[I] Inspect
REQ-061	Inactive Item Cursor Tag	All non-active menu items SHALL be prefixed with ` ` (two spaces at Column 0 and 1).	[I] Inspect
REQ-062	Menu Item 0 Text	Item 0 in the menu registry SHALL display `Check Tray` (Manual 1-pulse feed burst).	[I] Inspect
REQ-063	Menu Item 1 Dynamic Text	Item 1 SHALL dynamically render as `Stop Feeding` when `feedActive == true`, and `Start Feeding` when `feedActive == false`.	[D] Demo

ID	Title	Requirement Statement	Method
REQ-064	Menu Item 2 Text	Item 2 in the menu registry SHALL display `Feed Quantity`.	[I] Inspect
REQ-065	Menu Item 3 Text	Item 3 in the menu registry SHALL display `Feed Per Event`.	[I] Inspect
REQ-066	Menu Item 4 Text	Item 4 in the menu registry SHALL display `Start Time`.	[I] Inspect
REQ-067	Menu Item 5 Text	Item 5 in the menu registry SHALL display `End Time`.	[I] Inspect
REQ-068	Menu Item 6 Text	Item 6 in the menu registry SHALL display `Discharge Rate`.	[I] Inspect
REQ-069	Menu Item 7 Text	Item 7 in the menu registry SHALL display `Diagnostics`.	[I] Inspect
REQ-070	Menu Item 8 Text	Item 8 in the menu registry SHALL display `Set Real Clock`.	[I] Inspect
REQ-071	Menu Item 9 Text	Item 9 in the menu registry SHALL display `Network Info`.	[I] Inspect
REQ-072	Menu Item 10 Text	Item 10 in the menu registry SHALL display `Power Monitor`.	[I] Inspect
REQ-073	Empty Slot Blanking	When viewing the final page of menu items, unused LCD rows SHALL be overwritten with 20 space characters to clear stale artifacts.	[I] Inspect
REQ-074	Diagnostics Submenu Items	Selecting `Diagnostics` SHALL open a submenu containing: `Test Feed (30-120s)`, `Test Spreader (2s)`, `Test Loader (2s)`, `Test Hooter (1s)`, `Ping Gateway`, and `< Back`.	[D] Demo
REQ-075	Diagnostics Manual Test Safety	Any manual motor diagnostic test SHALL automatically timeout and de-energize relays after a hard limit of $\le 5.0\text{ seconds}$.	[T] Test

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Category 5: Settings, Limits & NVS Persistence (REQ-076 to REQ-095)

ID	Title	Requirement Statement	Method
REQ-076	Feed Quantity Range	The Daily Feed Quantity setting SHALL be bounded between a minimum of 0.5 kg and a maximum of 120.0 kg.	[T] Test
REQ-077	Feed Quantity Step Size	The Feed Quantity adjustment step size SHALL be exactly 0.5 kg per button event.	[T] Test
REQ-078	Feed Per Event Range	The Feed Per Event setting SHALL be bounded between a minimum of 50 g and a maximum of 1000 g.	[T] Test
REQ-079	Feed Per Event Step Size	The Feed Per Event adjustment step size SHALL be exactly 50 g per button event.	[T] Test
REQ-080	Start Hour Range	The Schedule Start Hour setting SHALL be bounded between 06 (6 AM) and 17 (5 PM).	[T] Test
REQ-081	Start Minute Range	The Schedule Start Minute setting SHALL be bounded between 00 and 59 in increments of 1 minute.	[T] Test
REQ-082	End Hour Range	The Schedule End Hour setting SHALL be bounded between 07 (7 AM) and 18 (6 PM).	[T] Test

ID	Title	Requirement Statement	Method
REQ-083	End Minute Range	The Schedule End Minute setting SHALL be bounded between 00 and 59 in increments of 1 minute.	[T] Test
REQ-084	Operational Daylight Enclosure	The firmware SHALL enforce that `startHour >= 6` and `(endHour < 18 || (endHour == 18 && endMinute == 0))`.	[T] Test
REQ-085	Chronological Ordering Rule	The firmware SHALL reject any configuration where $(\text{EndHour} \times 60 + \text{EndMin}) \le (\text{StartHour} \times 60 + \text{StartMin})$.	[T] Test
REQ-086	Discharge Rate Range	The Auger Discharge Rate calibration parameter SHALL be bounded between 20 g/s and 500 g/s. Default: 100 g/s.	[T] Test
REQ-087	Discharge Rate Step Size	The Discharge Rate adjustment step size SHALL be exactly 10 g/s per button event.	[T] Test
REQ-088	NVS Flash Wear Reduction	Configuration SHALL only be written to NVS when values have actually changed (`memcmp != 0`).	[T] Test
REQ-089	Atomic NVS Transaction	Configuration saves SHALL utilize Preferences API `putBytes()` with verification read-back to guarantee atomicity.	[T] Test
REQ-090	Total Events Deduction	The total events count SHALL be calculated as $\text{TotalEvents} = \lceil (\text{FeedQuantity} \times 1000) / \text{FeedPerEvent} \rceil$.	[A] Analysis
REQ-091	Overfill Window Prevention	If $\text{TotalEvents} \times t_{\text{event}} \ge (\text{EndMin} - \text{StartMin}) \times 60$, the system SHALL flag `Schedule Error: Exceeds Window` on LCD.	[T] Test
REQ-092	Settings Uplink Notification	Whenever local settings are saved via LCD, the system SHALL queue a `LoRaSettingsPayload` (MsgType `0x22`) uplink within $\le 2\text{ seconds}$.	[D] Demo
REQ-093	Clock Setting Field 0	In `EDIT_CLOCK` , Field 0 SHALL adjust Hours (`00` to `23`). Pressing SELECT moves active cursor to Field 1.	[D] Demo
REQ-094	Clock Setting Field 1	In `EDIT_CLOCK` , Field 1 SHALL adjust Minutes (`00` to `59`). Pressing SELECT commits time to DS3231 RTC.	[D] Demo
REQ-095	Factory Reset Key Combo	Holding `[BACK]` and `[SELECT]` simultaneously for $> 10\text{ seconds}$ at boot SHALL erase NVS and restore factory defaults.	[T] Test

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Category 6: Schedule Engine, Timing & Operational Window Constraints (REQ-096 to REQ-115)

ID	Title	Requirement Statement	Method
REQ-096	Strict Night Window Motor Lockdown	From `18:00:00` to `06:00:00 IST` , the FeedEngine SHALL enforce `state = IDLE` and lock Relays 1 and 2 de-energized.	[T] Test
REQ-097	Unarmed First Boot Invariant	On initial power-on or new flash, the feeder SHALL NOT start automated feeding until explicitly started by operator (`hasBeenRun == false`).	[T] Test
REQ-098	Night Arming Behavior	If `START FEEDING` is commanded between `18:00` and `06:00` , the engine SHALL set `hasBeenRun = true` , arm for `06:00 AM` , and remain in `STANDBY` .	[T] Test
REQ-099	Single Event Duration Formula	The duration of one pulse SHALL be calculated as $t_{\text{event}} = 3.0\text{s (Pre)} + (\text{FeedPerEvent} / \text{DischargeRate}) + 3.0\text{s (Post)}$.	[A] Analysis
REQ-100	Full Window Duration Formula	Total daylight window seconds SHALL be calculated as $t_{\text{window}} = (\text{EndHour} \times 3600 + \text{EndMin} \times 60) - (\text{StartHour} \times 3600 + \text{StartMin} \times 60)$.	[A] Analysis

ID	Title	Requirement Statement	Method
REQ-101	Base Gap Calculation	The nominal wait gap SHALL be calculated as $t_{\text{base_gap}} = \max(5, \lfloor (t_{\text{window}} - (\text{TotalEvents} \times t_{\text{event}})) / \text{TotalEvents} \rfloor)$.	[A] Analysis
REQ-102	Remaining Events Deduction	In active runs, remaining events SHALL be calculated dynamically as $N_{\text{rem}} = \max(0, \text{TotalEvents} - \text{CurrentEvent})$.	[A] Analysis
REQ-103	Dynamic Gap Recalculation	When recalculated mid-day, the gap SHALL be $t_{\text{dyn_gap}} = \max(5, \lfloor (t_{\text{rem_window}} - (N_{\text{rem}} \times t_{\text{event}})) / N_{\text{rem}} \rfloor)$.	[A] Analysis
REQ-104	Dynamic Gap Ceiling	If remaining time exceeds schedule requirements, $t_{\text{dyn_gap}}$ SHALL be capped such that no feed occurs past `EndHour:EndMinute`.	[T] Test
REQ-105	Minimum Inter-Pulse Gap	The system SHALL enforce an absolute minimum wait gap of $\ge 5\text{ seconds}$ between feed events to prevent motor thermal overload.	[T] Test
REQ-106	Day Rollover Autonomous Trigger	When $T_{\text{epoch}} / 86400 > \text{lastFedDay}$, the engine SHALL reset `currentEvent = 0`, `dispensedQuantity_g = 0.0`, and clear NVS run state.	[T] Test
REQ-107	Target Dispensed Weight Completion	When `dispensedQuantity_g >= FeedQuantity * 1000`, the engine SHALL set `state = FEED_FINISHED` and cease pulsing for the day.	[T] Test
REQ-108	Target Event Count Completion	When `currentEvent >= TotalEvents`, the engine SHALL set `state = FEED_FINISHED` and cease pulsing for the day.	[T] Test
REQ-109	End of Operational Hours Concession	When local time reaches `EndHour:EndMinute`, the engine SHALL transition to `FEED_FINISHED` regardless of uncompleted events.	[T] Test
REQ-110	Immediate First Pulse on Manual Start	When `START FEEDING` is triggered inside daylight hours (`06:00` to `18:00`), Event 1 SHALL fire within $\le 1.0\text{ second}$.	[D] Demo
REQ-111	Next Feed Epoch Projection	During `FC_WAIT`, `nextFeedEpoch` SHALL equal $T_{\text{pulse_end}} + t_{\text{gap}}$.	[A] Analysis
REQ-112	Telemetry Heartbeat in Gap Delay	During `FC_WAIT`, the system SHALL suppress rapid telemetry bursts and transmit periodic heartbeats at 60s intervals.	[T] Test
REQ-113	Time Drift Realignment	If the RTC time is adjusted mid-cycle by $> 60\text{ seconds}$, the engine SHALL immediately recalculate dynamic schedule math.	[T] Test
REQ-114	Completion Lockout on Same Day	Once `FEED_FINISHED` is reached on calendar day D, auto-start SHALL remain locked until calendar day $D+1$.	[T] Test
REQ-115	Check Tray Schedule Independence	Executing a manual `Check Tray` burst SHALL NOT increment `currentEvent` or alter the scheduled `nextFeedEpoch`.	[T] Test

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Category 7: 3-Phase Dispensing Engine & Actuator Control (REQ-116 to REQ-130)

ID	Title	Requirement Statement	Method
REQ-116	Phase 1 Spreader Pre-Warmup	Upon pulse trigger, the system SHALL assert GPIO 10 (Spreader Relay) HIGH for exactly $3.0\text{ seconds} \pm 50\text{ ms}$ while Loader remains LOW.	[T] Test
REQ-117	Phase 2 Loader Metering Engagement	At $T = 3.0\text{ seconds}$, the system SHALL assert GPIO 11 (Loader Auger Relay) HIGH while maintaining GPIO 10 HIGH.	[T] Test
REQ-118	Dispense Metering Duration Math	Loader Relay SHALL remain energized for exactly $t_{\text{meter}} = (\text{FeedPerEvent} / \text{DischargeRate})\text{ seconds} \pm 50\text{ ms}$.	[T] Test

ID	Title	Requirement Statement	Method
REQ-119	Phase 3 Loader Cutoff	At $T = 3.0 + t_{\text{meter}}$, the system SHALL de-energize GPIO 11 (Loader Relay LOW).	[T] Test
REQ-120	Phase 3 Spreader Post-Clear Duration	GPIO 10 (Spreader Relay) SHALL remain energized for exactly $3.0\text{ seconds} \pm 50\text{ ms}$ following Loader cutoff.	[T] Test
REQ-121	Total Actuation Relay De-energize	At $T = 3.0 + t_{\text{meter}} + 3.0$, the system SHALL de-energize GPIO 10 (Spreader Relay LOW), completing the pulse.	[T] Test
REQ-122	Cumulative Dispense Math Tracking	Upon completion of Phase 3, the engine SHALL increment `dispensedQuantity_g += FeedPerEvent` and `currentEvent += 1`.	[A] Analysis
REQ-123	NVS Run State Checkpoint	The system SHALL persist `currentEvent`, `dispensedQuantity_g`, and `lastFedDay` to NVS at the end of each pulse.	[T] Test
REQ-124	Dispense Telemetry Event Generation	Upon completion of Phase 3, the system SHALL immediately queue a `LoRaUplinkPayload` with MsgType `0x21` (Event Report).	[D] Demo
REQ-125	Relay Non-Overlap Failsafe	The firmware SHALL guarantee that the Loader Relay (GPIO 11) is NEVER energized while the Spreader Relay (GPIO 10) is de-energized.	[T] Test
REQ-126	Relay Command Inversion Guard	If GPIO 10 fails to assert within 50ms, GPIO 11 SHALL NOT be asserted, and `MOTOR FAULT` SHALL be raised.	[T] Test
REQ-127	Spreader Motor Current Sampling	During Phase 1, Phase 2, and Phase 3, motor current SHALL be measured via INA219 every 50 ms.	[T] Test
REQ-128	Active Pulse LED Indication	The STAT2 Green LED SHALL flash at 5 Hz during Phase 1, Phase 2, and Phase 3.	[D] Demo
REQ-129	Emergency Motor Abort Latency	An emergency stop signal from software, overcurrent, or proximity SHALL de-energize both relays within $\le 10\text{ ms}$.	[T] Test
REQ-130	Watchdog Refresh in Dispense Loop	The FreeRTOS feed task SHALL pet the hardware watchdog timer every $\le 100\text{ ms}$ throughout the dispensing sequence.	[T] Test

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Category 8: Safety Interlocks, Current Monitoring & Motor Trip (REQ-131 to REQ-145)

ID	Title	Requirement Statement	Method
REQ-131	Overcurrent Trip Threshold	If INA219 measured current exceeds $3500\text{ mA} (3.5\text{A})$ for $\ge 150\text{ ms}$, the system SHALL trigger an Overcurrent Motor Trip.	[T] Test
REQ-132	Immediate Actuator Cutoff on Trip	Upon Overcurrent Trip, GPIO 10 and GPIO 11 SHALL be forced LOW within $\le 5\text{ ms}$.	[T] Test
REQ-133	Acoustic Overcurrent Alarm	Upon Overcurrent Trip, the piezo hooter (GPIO 42) SHALL sound continuously for 3.0 seconds.	[D] Demo
REQ-134	Visual Overcurrent Alarm	Upon Overcurrent Trip, the STAT1 Red LED SHALL latch ON and the RGB LED SHALL illuminate solid RED (`255, 0, 0`).	[D] Demo
REQ-135	System State Latch on Trip	The system state SHALL transition to `SystemState::ERROR_FATAL` with error code `ERR_OVERCURRENT`.	[T] Test
REQ-136	Telemetry Bitmask Assertion on Trip	The telemetry engine SHALL assert `Bit 1` (`0x02` Overcurrent Trip) in `statusFlags` on all subsequent uplinks.	[T] Test
REQ-137	Cloud Alert Uplink on Trip	The system SHALL immediately transmit an emergency uplink packet with MsgType `0x04` (Feed Paused / Alarm).	[D] Demo

ID	Title	Requirement Statement	Method
REQ-138	LCD Motor Jam Notification	The LCD SHALL display `State: MOTOR TRIP` on Row 1 and `Overcurrent Check HW` on Row 2.	[I] Inspect
REQ-139	Jam Auto-Clear Prohibition	The system SHALL NOT automatically restart feeding following an Overcurrent Trip. Manual user intervention is mandatory.	[T] Test
REQ-140	Manual Jam Clearance Reset	The user SHALL only clear a Motor Trip by pressing `[BACK]` from the Home Screen or power-cycling the device.	[D] Demo
REQ-141	Brownout Voltage Floor	If battery voltage drops below 10.2V for $> 500\text{ ms}$, the system SHALL inhibit motor actuation and report `LOW BATTERY`.	[T] Test
REQ-142	Solar Overvoltage Protection	If voltage exceeds 16.0V, the system SHALL log a warning and clamp charging indicators.	[T] Test
REQ-143	Thermal Cutoff Protection	If DS3231 internal temperature sensor reports $> 70^\circ\text{C}$, motor dispense operations SHALL be paused until temperature $\le 60^\circ\text{C}$.	[T] Test
REQ-144	Relay Weld Detection	If current $> 100\text{ mA}$ is detected when both relays are commanded OFF, system SHALL latch `RELAY HARDWARE FAULT`.	[T] Test
REQ-145	Current Sensor Communication Loss	If INA219 fails to acknowledge on I2C during dispense, the system SHALL limit maximum pulse time to $2.0 \times t_{\text{meter}}$ as a failsafe.	[T] Test

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Category 9: Hopper Level Sensing, Proximity & Hooter Alarm (REQ-146 to REQ-160)

ID	Title	Requirement Statement	Method
REQ-146	Proximity Sensor Hardware Input	The hopper level sensor SHALL be monitored via GPIO 16 through a PC817 optocoupler with a 10kΩ pull-up.	[I] Inspect
REQ-147	Hopper Empty Sensor State	When feed level drops below the sensor window, the optocoupler transistor de-energizes, pulling GPIO 16 HIGH (Active HIGH dry).	[T] Test
REQ-148	Hopper Empty Debounce Window	GPIO 16 must remain continuously HIGH for $\ge 100\text{ ms}$ before triggering `FEED EMPTY` to eliminate pellet bounce noise.	[T] Test
REQ-149	Feed Empty Mid-Pulse Abort	If `FEED EMPTY` triggers during Phase 1, Phase 2, or Phase 3, the Loader Relay SHALL de-energize immediately, followed by a 3s Spreader clear.	[T] Test
REQ-150	System State on Empty	The system state SHALL transition to `SystemState::FEED_PAUSED`.	[T] Test
REQ-151	Telemetry Bitmask on Empty	The telemetry engine SHALL assert `Bit 3` (`0x08` Feed Hopper Empty) and `Bit 4` (`0x10` Feeder Paused) in `statusFlags`.	[T] Test
REQ-152	Hooter Empty Acoustic Pattern	While paused due to empty hopper, the piezo hooter SHALL emit three 200 ms chirps separated by 100 ms pauses, repeating every 15 seconds.	[D] Demo
REQ-153	Visual Empty Indicator	The STAT1 Red LED SHALL flash at 1 Hz while paused due to empty hopper.	[D] Demo
REQ-154	LCD Empty Screen Display	The LCD SHALL display `Error: FEED EMPTY` on Row 1 and `[SEL]: Resume` on Row 2 of the Paused Screen.	[I] Inspect
REQ-155	Proximity Active Low Config Option	If `config.proxAlarmHigh == false`, the active level logic SHALL be inverted to support normally-closed proximity switches.	[T] Test
REQ-156	Resume Sensor Interlock	Pressing `[SELECT]` on the Paused Screen SHALL verify `hal.isProximityTriggered() == false` before permitting resume.	[T] Test

ID	Title	Requirement Statement	Method
REQ-157	Refusal on Uncleared Sensor	If `[SELECT]` is pressed while sensor is still dry (hopper empty), the hooter SHALL emit a negative double-buzz and remain paused.	[D] Demo
REQ-158	Hooter Mute Capability	Pressing `[BACK]` while on the Paused Screen SHALL mute the periodic acoustic hooter while retaining visual error flags.	[D] Demo
REQ-159	Auto-Resume Prohibition	The feeder SHALL NOT automatically resume dispensing when pellets are refilled without operator confirmation (LCD SEL or Cloud Downlink).	[T] Test
REQ-160	Pellets Refilled Telemetry	When hopper is refilled and resumed, an uplink with Bit 3 (`0x08`) cleared SHALL be transmitted within $\le 2\text{ seconds}$.	[D] Demo

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Category 10: Pause, Resume Engine & Manual Feed (REQ-161 to REQ-175)

ID	Title	Requirement Statement	Method
REQ-161	Same-Window Resume Invariant	If paused and resumed within the SAME wait cycle ($t_{\text{pause}} < t_{\text{gap}}$), the engine SHALL NOT recalculate intervals and retain `nextFeedEpoch` .	[T] Test
REQ-162	Same-Window Countdown Continuity	In Scenario A (Same-Window Resume), the countdown on Row 2 SHALL resume from $(T_{\text{nextFeed}} - T_{\text{current}})$.	[A] Analysis
REQ-163	Extended Pause Threshold	If total paused duration exceeds t_{gap}, the system SHALL classify the event as Scenario B (Extended Pause).	[A] Analysis
REQ-164	Extended Pause Dynamic Recalculation	In Scenario B, the engine SHALL invoke `recalcDynamic()` , re-distributing remaining kilograms across remaining daylight minutes until `EndHour:EndMin` .	[A] Analysis
REQ-165	Immediate Pulse on Extended Resume	In Scenario B, the feeder SHALL fire a compensatory feed pulse within $\le 1.0\text{ second}$ of resume confirmation.	[D] Demo
REQ-166	Feed Session Cancel Action	Pressing `[BACK]` on the Paused Screen SHALL set `state = IDLE` , disable `feedActive` , and cancel all remaining pulses for the day.	[D] Demo
REQ-167	Manual Check Tray Invocation	Selecting `Check Tray` from `MENU_LIST` SHALL initiate a single 1-pulse feed burst regardless of current gap delay state.	[D] Demo
REQ-168	Check Tray Night Lockdown	The system SHALL reject `Check Tray` requests received between `18:00` and `06:00 IST` with error `Outside Feed Window` .	[T] Test
REQ-169	Check Tray Weight Non-Deduction	Pellets dispensed during `Check Tray` SHALL be tracked under `trayCheckGrams` and SHALL NOT deduct from the scheduled daily target.	[T] Test
REQ-170	Check Tray Telemetry Record	Upon `Check Tray` completion, a dedicated uplink packet with MsgType `0x0A` (Tray Checkin) SHALL be transmitted.	[D] Demo
REQ-171	Remote Cloud Pause Execution	Upon receiving Downlink Command `0xA1` (CMD_STOP_FEED), the feeder SHALL pause within $\le 500\text{ ms}$ and assert `state = FEED_PAUSED` .	[T] Test
REQ-172	Remote Cloud Resume Execution	Upon receiving Downlink Command `0xA2` (CMD_RESUME_FEED), the feeder SHALL execute resume logic identical to LCD SELECT.	[T] Test
REQ-173	Remote Cloud Start Execution	Upon receiving Downlink Command `0xA0` (CMD_START_FEED), the feeder SHALL start the daily feeding session.	[T] Test
REQ-174	Pause Duration Metric Tracking	The firmware SHALL log total cumulative pause duration per day in minutes to evaluate feeder availability metrics.	[A] Analysis

ID	Title	Requirement Statement	Method
REQ-175	Motor Pre-Pause State Recovery	If paused during Phase 2 (Dispense), the resume cycle SHALL first run a full 3s Pre-warmup before re-engaging the auger.	[T] Test

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Category 11: Real-Time Clock, UTC/IST & Time Synchronization (REQ-176 to REQ-185)

ID	Title	Requirement Statement	Method
REQ-176	Hardware RTC Architecture	The system SHALL maintain local real-time clock via an external DS3231 I2C RTC with integrated temperature-compensated crystal.	[I] Inspect
REQ-177	Local Timezone Representation	All time variables inside the DS3231 (`Hour` , `Minute` , `Second` , `Day` , `Month` , `Year`) SHALL represent **Indian Standard Time (IST, UTC+5:30)**.	[T] Test
REQ-178	Firmware Epoch Derivation	The firmware function `rtcMgr.getEpoch()` SHALL compute seconds from `1970-01-01` assuming stored RTC fields are local time components.	[A] Analysis
REQ-179	Calendar Day Index Formula	The calendar day index SHALL be calculated as $\text{CurrentDay} = \lfloor \text{LocalEpoch} / 86400 \rfloor$, guaranteeing midnight rollover at exactly 00:00:00 IST.	[A] Analysis
REQ-180	Software RTC Synchronization	The Software RTC fallback timer SHALL synchronize with DS3231 on every successful read to eliminate millisecond drift.	[T] Test
REQ-181	Cloud Time Synchronization Downlink	The cloud Lambda SHALL evaluate device clock on every uplink. If $	\text{DeviceEpoch} - \text{CloudEpoch}
REQ-182	RTC Adjustment Execution	Upon receiving Downlink `0xA4` (`CMD_SYNC_RTC`), the firmware SHALL write the received 32-bit epoch to DS3231 via `rtc.adjust()` .	[T] Test
REQ-183	RTC Fault Telemetry Flag	If the DS3231 oscillator stop flag (OSF) is asserted, the firmware SHALL assert `Bit 2` (`0x04` RTC Fault) in `statusFlags` .	[T] Test
REQ-184	Smooth Schedule Realignment	When RTC is updated by cloud downlink, ongoing wait delays SHALL NOT jump erratically; `nextFeedEpoch` SHALL be recomputed smoothly.	[T] Test
REQ-185	Daylight Savings Immunity	The firmware SHALL NOT implement daylight savings adjustments, operating on continuous UTC+5:30 year-round.	[I] Inspect

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Category 12: Wireless Telemetry & LoRaWAN Communications Contract (REQ-186 to REQ-200)

ID	Title	Requirement Statement	Method
REQ-186	Uplink Packet Size	The binary LoRaWAN uplink payload SHALL strictly measure **16 bytes** with `__attribute__((packed))` struct alignment.	[T] Test
REQ-187	Uplink Port Mapping	All telemetry uplink packets SHALL be transmitted on LoRaWAN **FPort 1**.	[T] Test
REQ-188	Downlink Port Mapping	All cloud commands and configuration downlinks SHALL be received on LoRaWAN **FPort 2**.	[T] Test
REQ-189	LoRaWAN Radio Modulation	Uplinks SHALL transmit in the IN865 band (865–867 MHz) using Spreading Factor SF7 to SF12, 125 kHz bandwidth.	[T] Test
REQ-190	Bit 5 Diagnostic Isolation	In the LoRa binary payload, `Bit 5` (`0x20`) of `statusFlags` SHALL strictly denote **SD Card Fault / Missing Card**.	[I] Inspect

ID	Title	Requirement Statement	Method
REQ-191	Cloud Misinterpretation Prohibition	Cloud ingestion services SHALL NOT interpret `Bit 5` (`0x20`) as feed active. `FEEDING` state SHALL strictly depend on message type and daylight window.	[I] Inspect
REQ-192	Day/Night LoRaWAN Class Switching	The system SHALL operate in **Class C** (Continuous RX) between 06:00 and 18:00, and switch to **Class A** (Power conservation) between 18:00 and 06:00.	[D] Demo
REQ-193	Downlink Command Packet Format	The binary downlink packet SHALL strictly measure **12 bytes**, containing `cmdType` , `feedQty_100g` , `fpe_g` , `shour` , `smin` , `ehour` , `emin` , and `rate` .	[T] Test
REQ-194	Downlink Execution Acknowledgment	Within $\le 5\text{ seconds}$ of executing a downlink command, the feeder SHALL transmit an uplink reflecting the new status flags.	[D] Demo
REQ-195	Uplink Queue Depth	The firmware telemetry queue SHALL hold a minimum of 20 unacknowledged events (`TELEMETRY_QUEUE_SIZE >= 20`) in FreeRTOS memory.	[T] Test
REQ-196	Telemetry Deduplication	The cloud DynamoDB ingestion Lambda SHALL deduplicate incoming uplinks using primary key `(DeviceId, Timestamp)` .	[T] Test
REQ-197	Daytime Telemetry Filtering	Cloud queries for "Today's Dispensed Feed" SHALL strictly filter telemetry records with `Timestamp >= today_start_epoch (00:00:00 IST)` .	[T] Test
REQ-198	Zero Item Rollover Invariant	When no telemetry records exist for the current calendar day, the API SHALL return `0.00 kg` and `0` events.	[T] Test
REQ-199	LoRa SX1262 SPI Transaction Mutex	All SPI bus interactions with SX1262 SHALL be protected by a FreeRTOS mutex to prevent collision with SD card transactions.	[T] Test
REQ-200	Link Quality Telemetry	The feeder SHALL record and uplink RSSI and SNR metrics to enable RF network health analytics on the farm dashboard.	[T] Test

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Category 13: Web & Mobile Dashboard UI Contract (REQ-201 to REQ-215)

ID	Title	Requirement Statement	Method
REQ-201	Zero Jargon Presentation	The dashboard SHALL display human-friendly state badges (`STANDBY` , `FEEDING` , `PAUSED` , `COMPLETED` , `OFFLINE`). Hex codes SHALL NOT be displayed.	[I] Inspect
REQ-202	Night Window Status Display	When local IST time is between 18:00 and 06:00, the dashboard status badge SHALL display **`STANDBY`** (never `FEEDING`).	[T] Test
REQ-203	Night Window Next Feed Display	Between 00:00 and 06:00 IST, the Next Feed card SHALL display **`06:00 AM`** with subtitle `Starts today at 06:00 AM` .	[I] Inspect
REQ-204	Evening Next Feed Display	Between 18:00 and 24:00 IST, the Next Feed card SHALL display **`06:00 AM`** with subtitle `Starts tomorrow at 06:00 AM` .	[I] Inspect
REQ-205	Pre-Dawn Progress Bar Integrity	Between 00:00 and 06:00 IST on a new calendar day, the central circular gauge SHALL display **`0 / 50 EVENTS TODAY (0.0%)`** .	[T] Test
REQ-206	Pre-Dawn Dispensed Weight Integrity	Between 00:00 and 06:00 IST, the Dispensed Today metric card SHALL display **`0.00 kg`** (never yesterday's accumulated weight).	[T] Test
REQ-207	Standby Action Button Mapping	When feeder is in `STANDBY` , the primary button SHALL display **`START FEEDING`** (enabled) and the secondary button SHALL display `STOP FEEDING` (disabled).	[D] Demo
REQ-208	Night Arming Operator Confirmation	Clicking `START FEEDING` during night hours SHALL display an informational prompt: "*Feeder armed for 06:00 AM. No feed will dispense tonight.*"	[D] Demo
REQ-209	Feeding Action Button Mapping	When feeder is in `FEEDING` , the primary button SHALL display **`FEED BURST NOW`** and secondary button SHALL display **`STOP FEEDING`** (active red).	[D] Demo

ID	Title	Requirement Statement	Method
REQ-210	Paused Action Button Mapping	When feeder is in `PAUSED`, the primary button SHALL display **`RESUME FEEDING`** (active cyan) and secondary button SHALL display **`STOP FEEDING`**.	[D] Demo
REQ-211	Completed Action Button Mapping	When feeder is in `COMPLETED`, the primary button SHALL display **`START FEEDING`** (armed for tomorrow) and secondary button SHALL be disabled.	[D] Demo
REQ-212	Mobile Viewport Responsiveness	The web dashboard layout SHALL render responsively on mobile viewports (375px to 430px width) without horizontal scrolling.	[D] Demo
REQ-213	Auto-Refresh Polling Cycle	The dashboard SHALL poll `GET /api/devices` every 5.0 seconds to reflect near-real-time farm telemetry.	[T] Test
REQ-214	History Tab Event Log Isolation	The History tab SHALL only list feed pulses timestamped on the active operational date. Past sessions SHALL be grouped under "Previous Sessions".	[T] Test
REQ-215	Cloud Command Pending Pill	While a downlink command is queued in DynamoDB awaiting delivery, the UI SHALL display a pulsing indicator: ⏳ `Command Sync Pending`.	[D] Demo

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4. Verification & Traceability Cross-Reference

System Tier	Requirements Covered	Primary Verification Tool
Firmware Core & HAL	REQ-001 to REQ-018, REQ-041 to REQ-058, REQ-116 to REQ-160	Hardware-in-the-loop (HIL), Oscilloscope, Logic Analyzer
20x4 LCD Display UI	REQ-019 to REQ-040, REQ-059 to REQ-075	Visual String Budget Audit, LCD Character Counter Scripts
Feed Engine & Math	REQ-076 to REQ-115, REQ-161 to REQ-185	Unit Tests, Mathematical Timing Analysis, Epoch Rollover Sim
LoRaWAN & Cloud Microservices	REQ-186 to REQ-200	AWS CloudWatch, Packet Logger, LoRaWAN Gateway Console
Web & Mobile Dashboard	REQ-201 to REQ-215	Selenium WebDriver, Mobile Emulation, API Integration Tests

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End of Specification

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